```python
import pandas as pd
import glob
import os

# List all CSVs
data_dir = 'data/raw/'  # Adjust if needed
csv_files = glob.glob(os.path.join(data_dir, '*.csv'))
print(f"Total files: {len(csv_files)}")
for f in sorted(csv_files):
    df = pd.read_csv(f)
    print(f"\n{f.split('/')[-1]}: {df.shape[0]} rows, {df.shape[1]} cols")
    print(df.columns.tolist()[:10], '...' if len(df.columns) > 10 else '')  # First 10 cols
    print(df.dtypes.value_counts())
    print("Missing %:", df.isnull().mean().mean())
```

```
Total files: 21

agents_pick_rates.csv: 11948 rows, 6 cols
['Tournament', 'Stage', 'Match Type', 'Map', 'Agent', 'Pick Rate']
str    6
Name: count, dtype: int64
Missing %: 0.0

draft_phase.csv: 864 rows, 7 cols
['Tournament', 'Stage', 'Match Type', 'Match Name', 'Team', 'Action', 'Map']
str    7
Name: count, dtype: int64
Missing %: 0.0

eco_rounds.csv: 12608 rows, 11 cols
['Tournament', 'Stage', 'Match Type', 'Match Name', 'Map', 'Round Number', 'Team', 'Load
str        9
int64      1
float64    1
Name: count, dtype: int64
Missing %: 0.09090909090909091

eco_stats.csv: 4090 rows, 9 cols
['Tournament', 'Stage', 'Match Type', 'Match Name', 'Map', 'Team', 'Type', 'Initiated',
str        7
float64    1
int64      1
Name: count, dtype: int64
Missing %: 0.022222222222222223

kills.csv: 30675 rows, 13 cols
['Tournament', 'Stage', 'Match Type', 'Match Name', 'Map', 'Player Team', 'Player', 'Ene
str        10
float64     3
Name: count, dtype: int64
Missing %: 0.09206695504984014

kills_stats.csv: 4090 rows, 20 cols
['Tournament', 'Stage', 'Match Type', 'Match Name', 'Map', 'Team', 'Player', 'Agents', '
float64    9
str        8
int64      3
Name: count, dtype: int64
Missing %: 0.3399877750611247

maps_played.csv: 379 rows, 5 cols
['Tournament', 'Stage', 'Match Type', 'Match Name', 'Map']
```

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str    5
Name: count, dtype: int64
Missing %: 0.0

maps_scores.csv: 379 rows, 16 cols
['Tournament', 'Stage', 'Match Type', 'Match Name', 'Map', 'Team A', 'Team A Score', 'Te
str        8
int64      6
float64    2
Name: count, dtype: int64
Missing %: 0.11411609498680739

maps_stats.csv: 412 rows, 7 cols
['Tournament', 'Stage', 'Match Type', 'Map', 'Total Maps Played', 'Attacker Side Win Per
str      6
int64    1
Name: count, dtype: int64
Missing %: 0.0

overview.csv: 15690 rows, 21 cols
['Tournament', 'Stage', 'Match Type', 'Match Name', 'Map', 'Player', 'Team', 'Agents', '
str        11
float64    10
Name: count, dtype: int64
Missing %: 0.0171112932107196

players_ids.csv: 244 rows, 2 cols
['Player', 'Player ID']
str      1
int64    1
Name: count, dtype: int64
Missing %: 0.0

players_stats.csv: 5231 rows, 25 cols
['Tournament', 'Stage', 'Match Type', 'Player', 'Teams', 'Agents', 'Rounds Played', 'Rat
str        10
int64       8
float64     7
Name: count, dtype: int64
Missing %: 0.0430357484228637

rounds_kills.csv: 26552 rows, 13 cols
['Tournament', 'Stage', 'Match Type', 'Match Name', 'Map', 'Round Number', 'Eliminator T
str      12
int64     1
Name: count, dtype: int64
Missing %: 0.0

scores.csv: 144 rows, 9 cols
['Tournament', 'Stage', 'Match Type', 'Match Name', 'Team A', 'Team B', 'Team A Score',
str      7
int64    2
Name: count, dtype: int64
Missing %: 0.0

team_mapping.csv: 48 rows, 2 cols
['Abbreviated', 'Full Name']
str    2
Name: count, dtype: int64
Missing %: 0.0

teams_ids.csv: 48 rows, 2 cols
```

```
['Team', 'Team ID']
str      1
int64    1
Name: count, dtype: int64
Missing %: 0.0

teams_picked_agents.csv: 5632 rows, 9 cols
['Tournament', 'Stage', 'Match Type', 'Map', 'Team', 'Agent', 'Total Wins By Map', 'Tota
str      6
int64    3
Name: count, dtype: int64
Missing %: 0.0

tournaments_stages_match_types_ids.csv: 77 rows, 6 cols
['Tournament', 'Tournament ID', 'Stage', 'Stage ID', 'Match Type', 'Match Type ID']
str        3
float64    2
int64      1
Name: count, dtype: int64
Missing %: 0.021645021645021644

tournaments_stages_matches_games_ids.csv: 379 rows, 9 cols
['Tournament', 'Tournament ID', 'Stage', 'Stage ID', 'Match Type', 'Match Name', 'Match 
str      5
int64    4
Name: count, dtype: int64
Missing %: 0.0

win_loss_methods_count.csv: 758 rows, 14 cols
['Tournament', 'Stage', 'Match Type', 'Match Name', 'Map', 'Team', 'Elimination', 'Deto
int64    8
str      6
Name: count, dtype: int64
Missing %: 0.0

win_loss_methods_round_number.csv: 16104 rows, 9 cols
['Tournament', 'Stage', 'Match Type', 'Match Name', 'Map', 'Round Number', 'Team', 'Meth
str      8
int64    1
Name: count, dtype: int64
Missing %: 0.0
```

Project Overview

The dataset contains 21 CSV files with Valorant Champions Tour (VCT) 2026 data across Kickoff (all regions) and Masters Santiago, including match results, player stats (like Average Combat Score, Kills to Death to Assist Ratio), agent pick rates/bans, rounds, economy, and maps.

These files were download from a bigger kaggle dataset: https://www.kaggle.com/datasets/ryanluong1/valorant-champion-tour-2021-2023-data

however, I am only using the relevent data from 2026, nothing prior to this year.

There is around 50,000-100,000 total observations across files (e.g., agents_pick_rates.csv: ~5,000 rows of pick instances). ~20-30 predictors per main table post-merge (e.g., ACS, FKPR, pick_rate, map, team).

I am working with both numerical varibles, such as in pick_rate (0-100), ACS/KDA (~1.1-1.3); and categorical variables such as TournemtnStage (e.g., "VCT 2026 Americas Kickoff), Map(Abyss, Bind), and Agents (Viper, Astra).

Missing data mainly is included in the Agents that were not used. So their pick rates are null.

My plan is to drop them if said value is under 5%.

Project Research Questions

I am interested in predicting the outcomes of upcoming VCT matches. For example Sentinels vs 100T. My response varible is binary. 1 if a team wins and 0 if they do not. This is all derived from matches.csv score differentials, indicating match outcome.

This is a classification model, since it as a binary outcome of win/loss. Some useful predictoes include average team ACS (>210 strong), K/D ration (+1.2) and even first kill per round.

The goal of my model is to be predictive in forecasting win probabilities for upcoming matches. I would like to also add descriptive elements in predicting ACS and other stats.

Project Timeline

As of now, I have my dataset loaded. The 21 csv files is more than enough data for me to go through in determining an accuate machine learning project. As of now until end of week 3 will be my data analysis and getting more familiar with python libraries that will be included during the project. After that it's all about building the project. Figure out how to actually make predictors and understand them is more important than just making some chart using old data. Calculating the attributes will help take into account the probabilties I am looking for.

Questions

As of now I do not have any further questions. If I do office hours are definetly my best friend and I already plan to go often even without having questions regarding the final project.